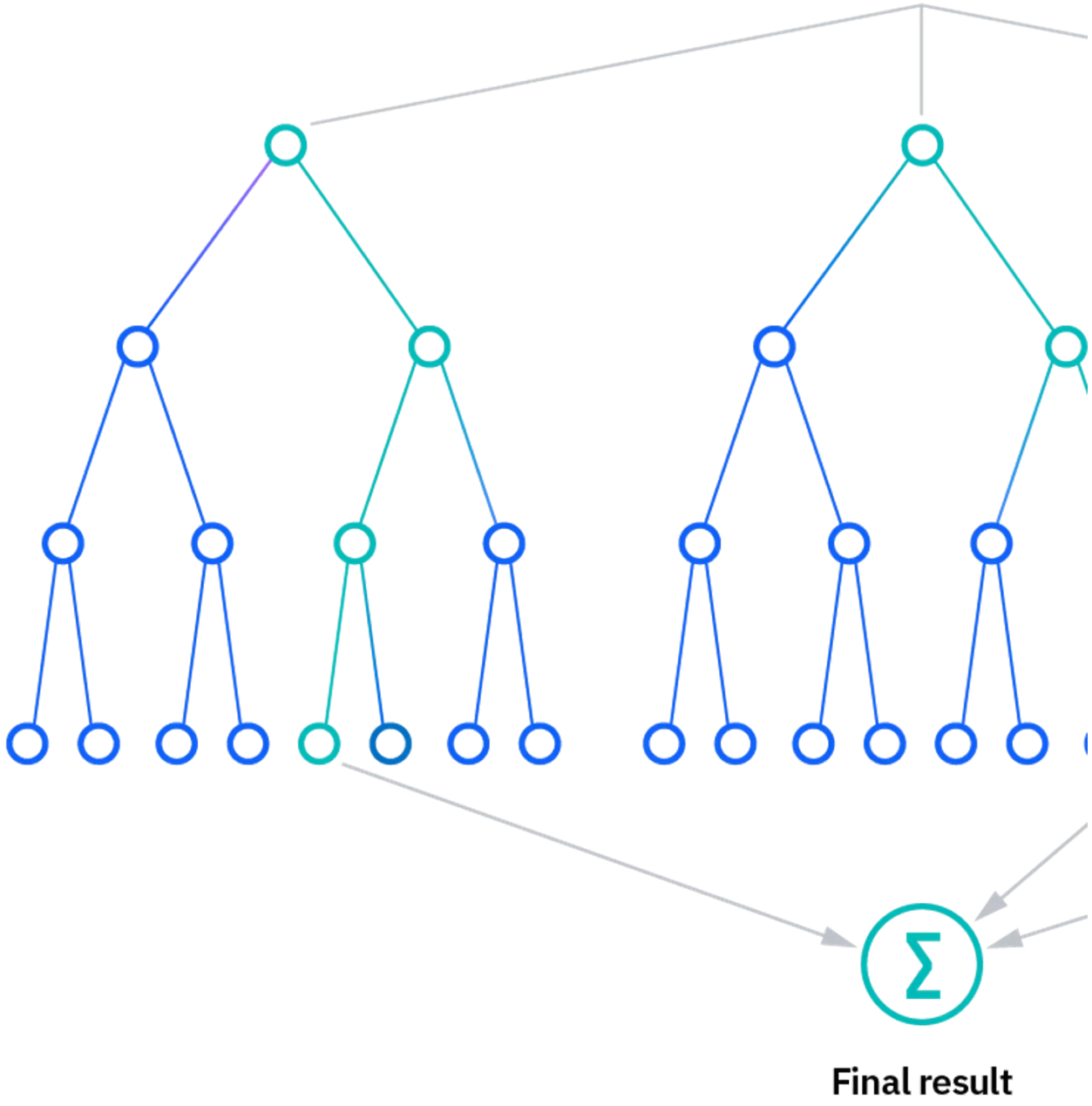
Random Forest

it is an ensemble learning technique
--> **improve the model accuracy and reduce the errors**

it combines the output of multiple decision tree to reach a single result.

it is an extension of the bagging method as it utilizes both bagging and feature randomness to create an uncorrelated forest of decision tree. (**feature randomness** = **feature bagging** = **the random subspace method**) generates a random subset of features, which ensures low correlation among decision trees



Assumptions

- Each tree makes its own decisions: every tree in the forest make its own predictions without relying on others
- Random Parts of the data are used: each tree is built using random samples and features to reduce mistakes
- Enough data is needed: sufficient data ensures the trees are different and learn unique patterns and variety
- Different predictions improve accuracy: combining the predictions from different trees leads to a more accurate final result.

Working of Random Forest Algorithm

- **Create Many Decision Trees:** The algorithm makes many decision trees each using a random part of the data. So every tree is a bit different.
- **Pick Random Features:** When building each tree it doesn't look at all the features (columns) at once. It picks a few at random to decide how to split the data. This helps the trees stay different from each other.
- **Each Tree Makes a Prediction:** Every tree gives its own answer or prediction based on what it learned from its part of the data.
- **Combine the Predictions:** For classification we choose a category as the final answer is the one that most trees agree on i.e majority voting and for regression we predict a number as the final answer is the average of all the trees predictions.
- **Why It Works Well**: Using random data and features for each tree helps avoid overfitting and makes the overall prediction more accurate and trustworthy.

Pros and Cons

Pros:

- Random Forest provides very accurate predictions even with large datasets.
- Random Forest can handle missing data well without compromising with accuracy.
- It doesn't require normalization or standardization on dataset.
- When we combine multiple decision trees it reduces the risk of overfitting of the model.
  Cons:
- It can be computationally expensive especially with a large number of trees.
- It's harder to interpret the model compared to simpler models like decision trees.